Table 3. Summary of catchment attributes. Climate indices and hydrological signatures were computed for the period 1 January 1990–31 December 2010. Index i refers to the precipitation product ($i = 1, 2, 3, 4$ for $\text{precip}_{\text{cr2met}}$, $\text{precip}_{\text{tmpa}}$, $\text{precip}_{\text{chirps}}$ and $\text{precip}_{\text{mswep}}$, respectively).

Attribute class	Attribute name	Description	Unit	Data source	Reference
Location and topography	gauge_id	catchment identifier (corresponds to the station code provided by DGA)	–	Gauges information collected from http://explorador.cr2.cl (last access: March 2018)	Sect. 3.1
	gauge_name	gauge name (based on DGA records)	–		
	gauge_lat	gauge latitude (based on DGA records)	° S		
	gauge_lon	gauge longitude (based on DGA records)	° W		
	gauge_record_start	start date of streamflow records	–		
	gauge_record_end	end date of streamflow records	–		
	gauge_n_obs	number of days with valid streamflow records	day		
	area	catchment area	km^2	ASTER GDEM 30 m raster data (Tachikawa et al., 2011)	
	elev_gauge	gauge elevation (catchment outlet) obtained from the 30 m ASTER GDEM elevation data and the location provided by DGA	m a.s.l.		
	elev_mean	catchment mean elevation	m a.s.l.		
	elev_med	catchment median elevation	m a.s.l.		
	elev_max	catchment maximum elevation	m a.s.l.		
	elev_min	catchment minimum elevation	m a.s.l.		
	slope_mean	catchment mean slope	$m\,km^{-1}$		
	nested_inner	number of inner catchments contained within gauge_id catchment (the gauge_id for the inner catchments can be obtained from the hierarchy matrix described in Sect. 3.1.1)	–	–	
	nested_outer	number of catchments containing gauge_id catchment (the gauge_id for the outer catchments can be obtained from the hierarchy matrix described in Sect. 3.1.1)	–		
	location_type	classification in "coastal (or low elevation)", "foothill" and "altiplano" catchments, based on gauge elevations (gauge_elev) below 50 m a.s.l., between 1000 and 1200 m a.s.l., and above 3500 m a.s.l., respectively.	–	–	Sect. 3.2
Geological characteristics	geol_class_1st	most common geologic class in the catchment	–	Global Lithological Map database (GLiM) (Hartmann and Moosdorf, 2012)	Table 6 in A17
	geol_class_1st_frac	fraction of the catchment area associated with its most common geologic class	–		
	geol_class_2nd	second most common geologic class in the catchment	–		
	geol_class_2nd_frac	fraction of the catchment area associated with its second most common geologic class	–		
	carb_rocks_frac	fraction of the catchment area characterised as "carbonate sedimentary rocks"	–		
Land cover characteristics	crop_frac	Percentage of the catchment covered by croplands, level 1. Includes five types of level 2 classes: rice fields; greenhouse farming; other croplands; orchards; and bare croplands	%	30 m resolution land cover map provided by Zhao et al. (2016)	Sects. 3.1.3 and 3.2.3
	nf_frac	Percentage of the catchment covered by forest (level 1) classified as natural broadleaf (level 2) or natural conifer (level 2).	%		
	fp_frac	Percentage of the catchment covered by forest (level 1) classified as broadleaf plantations (level 2) or conifer plantations (level 2).	%		
	grass_frac	Percentage of the catchment covered by grasslands, level 1. Includes three types of level 2 classes: pastures; other grasslands; and withered grasslands.	%		
	shrub_frac	Percentage of the catchment covered by shrublands, level 1. Includes five types of level 2 classes: shrublands; shrubs and sparse trees mosaic; succulents; shrub plantations; and withered shrublands.	%		
	wet_frac	Percentage of the catchment covered by wetlands and water bodies (level 1). Includes six types of level 2 classes: marshlands; mudflats; other wetlands; lakes; reservoirs/ponds; rivers; and ocean.	%		
	imp_frac	Percentage of the catchment covered by impervious surfaces (level 1). Urbanised areas are usually contained in this class.	%		

Table 3. Continued.

Attribute class	Attribute name	Description	Unit	Data source	Reference
	barren_frac	Percentage of the catchment covered by barren lands (level 1). Includes three types of level 2 classes: dry salt flats; sandy areas; and bare exposed rocks	%		
	snow_frac	Percentage of the catchment covered by snow and ice, level 1. Includes two types of level 2 classes: snow and ice.	%		
	fp_nf_index	forest plantation index: calculated as the ratio between fp_frac and the total forested area (fp_frac + nf_frac)	–		
	forest_frac	fraction of the catchment covered by forests, including native forest and forest plantation (fp_frac + nf_frac)	%		
	dom_land_cover	dominant land cover class	–		
	dom_land_cover_frac	fraction of the basin associated with dominant land cover class	%		
	land_cover_missing	percentage of the basin not covered by the land cover map	%		
	glaciers_area	glacierised area within the catchment	km^2	Randolph Glacier Inventory v. 6.0 (RGI Consortium, 2017)	Sects. 3.1.4 and 3.2.3
	glaciers_frac	percentage of the catchment covered by glaciers.	%		
Climatic indices (computed for 1 April 1990 to 31 March 2010)	p_mean_i	mean daily precipitation of product *i*	$mm\,day^{-1}$	Precipitation, temperature and potential evapotranspiration products introduced in Sects. 3.1.6, 3.1.7 and 3.1.8, respectively.	Table 2 in A17
	p_mean_spread	coefficient of variation of basin-averaged mean annual precipitation from different products (standard deviation of p_mean_i from the four precipitation products, normalised by multi-product mean)	–		
	pet_mean	mean daily PET of pet_{har} product	$mm\,day^{-1}$		
	aridity_i	aridity, calculated as the ratio of mean daily PET (pet_mean) to mean daily precipitation (p_mean_i)	–		
	p_seasonality_i	seasonality and timing of precipitation (product *i*) estimated using sine curves to represent the annual temperature and precipitation cycles; positive (negative) values indicate that precipitation peaks in summer (winter); values close to 0 indicate uniform precipitation throughout the year)	–		
	frac_snow_i	fraction of precipitation (product *i*) falling as snow (i.e. on days colder than 0 °C)	–		
	high_prec_freq_i	frequency of high-precipitation days (≥ 5 times mean daily precipitation) for product *i*	$days\,yr^{-1}$		
	high_prec_dur_i	average duration of high-precipitation events (number of consecutive days ≥ 5 times mean daily precipitation), for product *i*	days		
	high_prec_timing_i	season during which most high-precipitation days (≥ 5 times mean daily precipitation) occur	season		
	low_prec_freq_i	frequency of dry days (< 1 $mm\,day^{-1}$), for product *i*	$day\,yr^{-1}$		
	low_prec_dur_i	average duration of dry periods (number of consecutive days < 1 $mm\,day^{-1}$), for product *i*	days		
	low_prec_timing_i	season during which most dry days (< 1 $mm\,day^{-1}$) occur, for product *i*	season		
Hydrological signatures (computed for 1 April 1990 to 31 March 2010)	q_mean	mean daily discharge	$mm\,day^{-1}$	Streamflow records collected from http://explorador.cr2.cl (last access: March 2018)	Table 3 in A17
	runoff_ratio_i	runoff ratio (ratio of mean daily discharge to mean daily precipitation), for product *i*	–		
	stream_elas_i	streamflow precipitation elasticity (sensitivity of streamflow to changes in precipitation at the annual timescale, using the mean daily discharge as reference and precipitation product *i*)	–		
	slope_fdc	slope of the flow duration curve, FDC (between the log-transformed 33rd and 66th streamflow percentiles)	–		
	baseflow_index	baseflow index (ratio of mean daily baseflow to mean daily discharge, hydrograph separation performed using the Ladson et al. (2013) digital filter with α set to 0.925)	–		
	hdf_mean	mean half-flow date (date on which the cumulative discharge since 1 April reaches half of the annual discharge)	day of the year		
	Q_5	5 % flow quantile (low flows)	$mm\,day^{-1}$		
	Q_{95}	95 % flow quantile (high flows)	$mm\,day^{-1}$		
	high_q_freq	frequency of high-flow days (> 9 times the median daily flow)	$day\,yr^{-1}$		

Table 3. Continued.

Attribute class	Attribute name	Description	Unit	Data source	Reference
	high_q_dur	average duration of high-flow events (number of consecutive days > 9 times the median daily flow)	days		
	low_q_freq	frequency of low-flow days (< 0.2 times the mean daily flow)	day yr^{-1}		
	low_q_dur	average duration of low-flow events (number of consecutive days < 0.2 times the mean daily flow)	days		
	zero_q_freq	percentage of days with $Q = 0$	%		
	swe_ratio	ratio of peak of snow water equivalent to mean annual discharge	–	SWE product developed by Cortés and Margulis (2017)	
Intervention	sur_rights_n	total number of granted surface rights within the catchment	–	Water Atlas developed by the DGA (2016a)	Sects. 3.1.10 and 3.2.6
	sur_rights_flow	annual flow calculated for consumptive permanent continuous surface water rights	$m^3 s^{-1}$		
	interv_degree	intervention degree defined as the annual flow of surface water rights (consumptive permanent continuous), normalised by mean annual streamflow.	–		
	gw_rights_n	total number of granted groundwater rights within the catchment	–		
	gw_rights_flow	annual flow calculated for consumptive permanent continuous groundwater water rights	$m^3 s^{-1}$		
	large_dam	0 if there is no dam within the catchment, 1 if there is at least one dam classified as "large"	–	http://www.ide.cl/descarga/capas/item/embalses-2016.html (last access: September 2017)	

The climatic attributes presented in Figs. 4 and 9 reveal basic features of Chilean climatology, described in more detail by Miller (1976) and Garreaud et al. (2017), among others. Mean annual precipitation ranges from less than 10 mm yr^{-1} in the Atacama Desert (northern Chile) to more than 3000 mm yr^{-1} in western Patagonia (Fig. 4b). Such a marked precipitation gradient reflects the relative influence of the subtropical, semi-permanent south-eastern Pacific anticyclone, and the frequent incursion of frontal systems at higher latitudes. The frequency of high-precipitation events also increases southward, with a maximum in southern–central Chile (Fig. 9d). The Andean domain in the Far North (Chilean Altiplano), receiving about 300 mm yr^{-1} above 4000 m a.s.l., is influenced by the monsoonal regime developing over the interior of the continent. In addition to the N–S gradient, precipitation increases strongly in the west–east direction due to the orographic enhancement of air masses over the windward slope of the Andes Cordillera (a factor of 2–3 between lowlands and windward slopes; Viale and Garreaud, 2014). PET has a more restricted range than precipitation (400–1400 mm yr^{-1}; Fig. 4d); therefore, the aridity index (PET/P, Fig. 9c) is higher in northern Chile (> 1.0) compared to that of southern regions (< 1.0). A positive precipitation seasonality (Fig. 9a) in northern Chile indicates precipitation peaks during summer (DJF), following the monsoonal precipitation regime governing in this region (Fig. 9f). In contrast, the negative seasonality values obtained for all macro-zones, except the Far North and Southern Patagonia, illustrate the increased storm frequency and high-precipitation events in most of the country during the winter (JJA) (Fig. 9a). Seasonality values close to zero indicate uniform precipitation throughout the year in Southern Patagonia (Fig. 9a and f). The zero-temperature isotherm during winter storms ranges between 1500 and 4000 m a.s.l., so most of the precipitation is liquid along the coast and interior valleys (Fig. 9b), while a larger fraction of solid precipitation is obtained in high-elevation basins.

3.2.5 Hydrological signatures

Hydrological signatures were computed for the period 1 April 1990 to 31 March 2010, as in Sect. 3.2.4. To exclude the effects of anthropic intervention on hydrologic response, we selected 94 catchments with valid daily streamflow records in at least 85 % of the period, based on the following criteria: interv_degree lower than 0.1 (i.e. less than 10 % of the annual streamflow allocated to surface rights), large_dam equal to zero (absence of large dams within the catchment), imp_frac lower than 5 % (negligible urban areas), copr_frac lower than 20 % (negligible irrigation effects) and fp_frac lower than 20 % (negligible forest plantation effects). Further, we excluded glacier-dominated catchments by selecting glacier_frac lower than 5 %. It should be noted that, despite calculating hydrological indices for a subset of catchments, raw daily time series for all 516 catchments are